

Closed-form expressions for computing flexoelectric coefficients in textured polycrystalline dielectrics

J.L. Buroni ^{a,b}, R. Melnik ^c, L. Rodríguez-Tembleque ^d, A. Sáez ^d, F.C. Buroni ^{b,e,*}

^a Department of Systems, Universidad CAECE sede Mar del Plata, Olavarría 2464, B7600ELH, Mar del Plata, Argentina

^b Department of Mechanical Engineering and Manufacturing, Escuela Técnica Superior de Ingeniería, Universidad de Sevilla, Camino de los Descubrimientos s/n, Seville E-41092, Spain

^c MS2Discovery Interdisciplinary Research Institute, Wilfrid Laurier University, 75 University Ave W, Waterloo, Ontario, N2L 3C5, Canada

^d Department of Continuum Mechanics and Structures, Escuela Técnica Superior de Ingeniería, Universidad de Sevilla, Camino de los Descubrimientos s/n, 41092, Seville, Spain

^e Department of Mechanical Engineering and Manufacturing, Escuela Politécnica Superior, Universidad de Sevilla, Calle Virgen de África 7, 41011, Seville, Spain

ARTICLE INFO

Keywords:

Voigt average
Crystallographic and texture symmetry
Barium titanate polycrystals
Wigner-D functions
Orientation distribution function
Isotropic condition

ABSTRACT

The volume average of the constituting properties of a textured polycrystalline aggregate (the Voigt average) provides a simple way to estimate the effective properties of such an aggregate in terms of its crystallographic symmetry, texture symmetry, and the properties of the constituting crystallites. Despite the interest in estimate or characterizing flexoelectric properties, few studies have addressed the problem of finding effective flexoelectric properties from the continuum phases of the constituents. Instead, most of them focus on single-crystals at the atomic level. This study provides, for the first time, closed-form expressions that allow the direct flexoelectric coefficients to be computed for textured polycrystalline dielectrics. Crystallites can belong to any class of cubic symmetry or to the tetragonal T_1 Laue group, while no restriction is imposed on texture symmetry. These expressions are reduced to very simple ones when cubic crystallites with fiber texture are considered. The proposed formulation provides information on the matrix structure of the homogenized material related to the crystallographic symmetry and texture. We close with some illustrative examples.

1. Introduction

The flexoelectric effect, which refers to the generation of an electric polarization in response to the gradient of the applied mechanical strain, has garnered significant attention in recent years, primarily due to its promising applications in energy harvesting and electromechanical sensing/actuating technologies, utilizing both piezoelectric and nonpiezoelectric materials [1,2]. In this context, it is crucial to recognize the foundational contributions of Raymond D. Mindlin in the realms of strain-gradient elasticity and the mathematical theory of piezoelectricity, which have significantly influenced contemporary studies in this field [3,4]. For further background, the recent study by Awad et al. presents a unified dynamical model in flexoelectricity, offering a glimpse into the historical developments in this area [5].

* Corresponding author at: Department of Mechanical Engineering and Manufacturing, Escuela Técnica Superior de Ingeniería, Universidad de Sevilla, Camino de los Descubrimientos s/n, Seville E-41092, Spain.

E-mail address: fburoni@us.es (F.C. Buroni).

<https://doi.org/10.1016/j.apm.2023.09.032>

Received 29 June 2023; Received in revised form 22 September 2023; Accepted 27 September 2023

Available online 5 October 2023

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While the flexoelectric effect has been previously explored in single-crystal materials at the atomic level [6,7], the accurate prediction and measurement of the flexoelectric response in textured polycrystalline materials continue to pose a challenging task, due to the intricate nature of their microstructures and the prevailing influence of size effects.

The field of micromechanics offers essential methods for characterizing the effective properties of heterogeneous materials in the context of continuum mechanics. Several techniques have been developed to calculate effective properties, including various averaging schemes within analytical micromechanics or theory of homogenization through numerical computations. The simplest way to estimate effective properties is to compute the volume average of the constituting properties, which is known as the Voigt average. In some cases, this can be performed analytically, which has significant implications for the design and optimization of devices.

In this regard, it is pertinent to acknowledge the contributions of Man and Huang, who have made significant strides in this field. In their 2011 study, they presented a simple explicit formula for the Voigt-Reuss-Hill average of elastic polycrystals with arbitrary crystal and texture symmetries, a method found to be comparably accurate to more sophisticated techniques such as self-consistent schemes [8]. Furthermore, their 2012 publication elucidated a representation theorem for material tensors of weakly-textured polycrystals, focusing on applications in elasticity [9]. While these works offer valuable insights and methodologies, it is worth noting that their scope is primarily confined to the analysis of elastic properties through fourth-order tensors with major and minor symmetries, and are particularly applicable in the context of weak anisotropy. These conditions delineate a certain boundary in their applicability, which contrasts with the broader scope and the different tensorial properties addressed in our study.

In the context of flexoelectricity in composites, several notable contributions have been made in the areas of computational homogenization and analytical frameworks. These include the works of Yvonnet et al. [10,11], Mawassay et al. [12], and Sidhardh et al. [13], among others. Additionally, the research of Guinovart-Sanjuán et al. [14] is noteworthy for providing simple closed-form formulas for the effective flexoelectricity of stratified multilayered composites with perfect interfaces. The availability of closed-form expressions for the relationship between the macro- and micro-scale characteristics can be used as a basis for tuning the macroscopic behavior by optimizing the statistics and properties of the constituents.

In light of these considerations, while the formulations presented in references [8,9] are adept for polycrystals, they encounter limitations when applied to flexoelectrics exhibiting strong anisotropy. This gap is adeptly bridged by the recent work of Buroni and Buroni [15], which introduces a general formulation capable of accommodating tensors of any rank and symmetry, thereby extending its applicability to the domain of flexoelectricity, as substantiated in their study [15].

Crystals belonging to the family of perovskites, with the general chemical formula ABO_3 , exhibit flexoelectric, piezoelectric, and pyroelectric behavior and dielectric permittivity higher than those of the usual dielectric materials. Examination of these crystallites is of significant importance because of their ability to undergo structural phase transformations, which makes them particularly useful in the production of advanced materials with diverse functionalities. Barium titanate (BTO) is a particularly notable example, since it is both a lead-free material [16] and environmentally friendly in its processing [17], while also exhibiting interesting phase transitions at different temperatures. At high temperatures, BTO exhibits the symmetry of a paraelectric cubic crystal belonging to the $\mathcal{O}_h$ point group. However, when cooled below 120°C but above 5°C , BTO becomes ferroelectric and transforms into a tetragonal crystal belonging to the C_{4v} point group [18]. Therefore, the study of tetragonal and cubic crystallites is crucial for the advancement of materials science and the development of innovative technologies.

The aim of this work is to provide fully analytical expressions for the Voigt average in polycrystals as an estimation of the effective direct flexoelectric tensor of aggregates of cubic or tetragonal crystals and general texture. Crystallites can belong to any class of cubic symmetry or to the tetragonal T_1 Laue group. Note that the C_{4v} point group, to which belongs the tetragonal BTO, it is included in the T_1 Laue group.

This article is organized as follows: In Section 2, we present the theory needed to obtain our formulas. We study in detail the symmetries involved. First, the symmetries of the eight-order tensor which governs how direct flexoelectric tensors transform, are derived. Then, the empirical Neumann principle [19] is used to derive the flexoelectric symmetries for cubic and tetragonal crystals through the corresponding covering operations. The isotropy condition is also presented. After that, the theory for orientational averages based on Wigner-D functions is presented. In Section 3 the closed-form expressions for the Voigt averages of polycrystals composed of cubic and tetragonal crystallites are derived. Due to practical interest, the reduction to cubic crystallographic symmetry and fiber texture symmetry are also presented. Some numerical results are presented in Section 4, and we close with the conclusions of the work in the last section.

In this work, the repeated index convention for addition is used. Explicit sums are indicated in expressions when convenient for the sake of clarity. Latin indices range from 1 to 3, while Greek indices range from 1 to 18.

2. Theory

2.1. Symmetries of the flexoelectric tensor

Assuming a linear piezoelectric and flexoelectric behavior, when a dielectric non-centrosymmetric material is subjected to small deformations, the electric polarization vector $\mathbf{P}$ is related to the strain tensor $\mathbf{S}$ and to its gradient, by the linear relation [1]

$$P_i = d_{ijk} S_{jk} + \mu_{ijkl} \frac{\partial S_{jk}}{\partial x_l}, \quad (1)$$

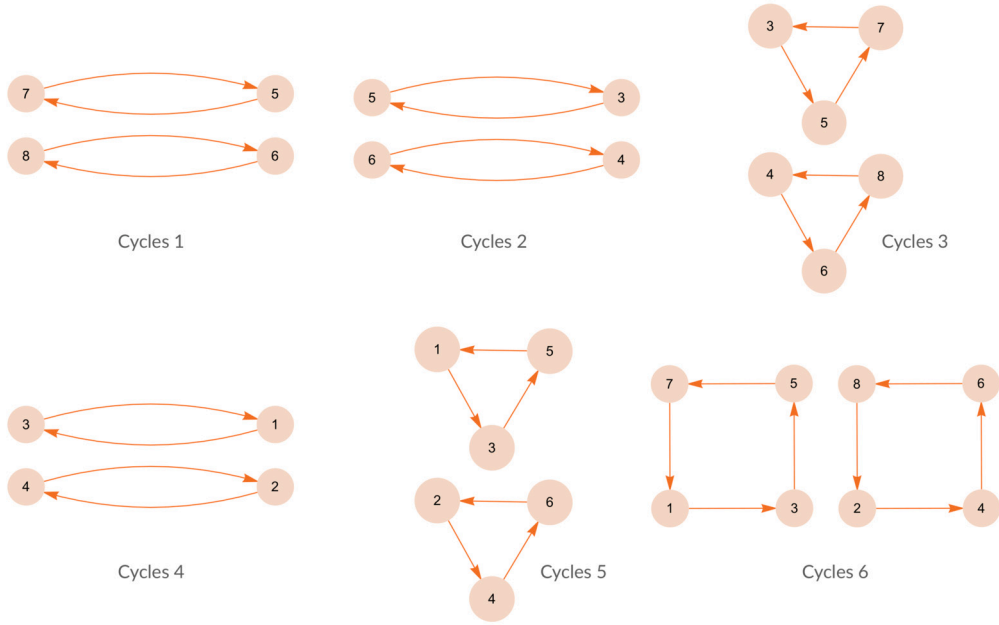


Fig. 1. Graphical representation of the permutations of the disjoint cycles of indices of the matrix $R^{\otimes 4}_{i_1 i_2 i_3 i_4 i_5 i_6 i_7 i_8}$.

where d_{ijk} are the components of the third-order piezoelectric tensor $\mathbf{d}$, and μ_{ijkl} , the components of the fourth-order direct flexoelectric tensor $\boldsymbol{\mu}$. If the material exhibits centrosymmetry, an odd-order tensor, such as the third-order piezoelectric tensor $\mathbf{d}$ vanishes, and the constitutive law (1) reduces to the tensorial relation

$$P_i = \mu_{ijkl} \frac{\partial S_{jk}}{\partial x_l}. \quad (2)$$

In a previous work [20], the authors have obtained closed-form expressions for the calculation of the Voigt average of textured piezoelectric aggregates of crystallites with any symmetry class. In the following, we focus our attention exclusively on the flexoelectric constitutive law (2), independently of whether the material is centrosymmetric or not.

If any skew-symmetric part exists in $\boldsymbol{\mu}$, the sum of its product with the symmetric strain tensor in (2) it would vanish, so we consider the flexoelectric tensor symmetrized by indices j and k . In other words, due to the symmetry $S_{ij} = S_{ji}$ of the strain tensor, the components of $\boldsymbol{\mu}$ satisfy the symmetry [21]

$$\mu_{ijkl} = \mu_{ikjl}. \quad (3)$$

Thus, for a dielectric material with triclinic symmetry, the number of independent flexoelectric coefficients is 54, in contrast to the well-known 21 independent coefficients for triclinic elasticities that arrive from their major and minor symmetries.

Consider a flexoelectric tensor $\boldsymbol{\mu}'$ referred to, for convenience, a principal material system. The components of such a tensor $\boldsymbol{\mu}'$ fixed to a crystallite that has some arbitrary orientation can be expressed in the global coordinate system x_i ($i = 1, 2, 3$) according to the transformation law for fourth-order tensors [22]

$$\boldsymbol{\mu} = \mathbf{R}^{\otimes 4} \boldsymbol{\mu}', \quad (4)$$

where $\mathbf{R}^{\otimes 4}$ is the eighth-order tensor which condenses the quadruple tensor product of the proper orthogonal matrix $\mathbf{R} \in SO(3)$, the special orthogonal group in three dimensions. The tensor $\mathbf{R}^{\otimes 4}$ exhibits specific symmetries, which are characterized by the permutations of the disjoint cycles of indices as schematically represented in Fig. 1. The symmetry relations can be expressed as follows:

$$\begin{aligned} R^{\otimes 4}_{i_1 i_2 i_3 i_4 i_5 i_6 i_7 i_8} &= R^{\otimes 4}_{i_1 i_2 i_3 i_4 i_7 i_8 i_5 i_6} = R^{\otimes 4}_{i_1 i_2 i_5 i_6 i_3 i_4 i_7 i_8} = R^{\otimes 4}_{i_1 i_2 i_7 i_8 i_3 i_4 i_5 i_6} \\ &= R^{\otimes 4}_{i_3 i_4 i_1 i_2 i_5 i_6 i_7 i_8} = R^{\otimes 4}_{i_5 i_6 i_1 i_2 i_3 i_4 i_7 i_8} = R^{\otimes 4}_{i_7 i_8 i_1 i_2 i_3 i_4 i_5 i_6}, \end{aligned} \quad (5)$$

with subindices ranging from 1 to 3. These relations reveal that out of the 6561 components of $\mathbf{R}^{\otimes 4}$, only 1545 are distinct, a feature that significantly influences the derivation of expressions for the Voigt average, as will be demonstrated in the subsequent sections.

For comparison purposes with the literature, we next introduce a 3×18 matrix representation for the tensor $\boldsymbol{\mu}$. Following Shu et al. [21], we adopt the reduced suffix notation for the gradient of the infinitesimal strain tensor $\frac{\partial S_{jk}}{\partial x_l} = \omega_{jkl}$ shown in the Table 1. Then, the operations involved in Eq. (2) can be written as

Table 1
Suffix notation correspondences between $\{i, j, k\}$ and α .

$\{i, j, k\} \mapsto \alpha$		
$\{1, 1, 1\} \mapsto 1$	$\{3, 3, 1\} \mapsto 7$	$\{1, 3, 1\} \mapsto 13$
$\{1, 1, 2\} \mapsto 2$	$\{3, 3, 2\} \mapsto 8$	$\{1, 3, 2\} \mapsto 14$
$\{1, 1, 3\} \mapsto 3$	$\{3, 3, 3\} \mapsto 9$	$\{1, 3, 3\} \mapsto 15$
$\{2, 2, 1\} \mapsto 4$	$\{1, 2, 1\} \mapsto 10$	$\{2, 3, 1\} \mapsto 16$
$\{2, 2, 2\} \mapsto 5$	$\{1, 2, 2\} \mapsto 11$	$\{2, 3, 2\} \mapsto 17$
$\{2, 2, 3\} \mapsto 6$	$\{1, 2, 3\} \mapsto 12$	$\{2, 3, 3\} \mapsto 18$

$$P_i = \mu_{i\alpha} \omega_\alpha, \quad (6)$$

where latin index range from 1 to 18, and

$$\omega_\alpha = \begin{cases} \omega_{ijk} \text{ with mapping } \{i, j, k\} \mapsto \alpha; & \text{if } \alpha < 10 \\ 2\omega_{ijk} \text{ with mapping } \{i, j, k\} \mapsto \alpha; & \text{if } \alpha \geq 10 \end{cases}, \quad (7)$$

resembling Voigt notation for elasticity or piezoelectricity. Explicitly, the 3×18 matrix in terms of the 54 components of the fourth-order direct flexoelectric tensor is

$$\mu_{3 \times 18} = \begin{pmatrix} \mu_{1111} & \mu_{1112} & \mu_{1113} & \mu_{1121} & \mu_{1122} & \mu_{1123} & \mu_{1131} & \mu_{1132} & \mu_{1133} & \mu_{1121} & \mu_{1122} & \mu_{1123} & \mu_{1131} & \mu_{1132} & \mu_{1133} & \mu_{1231} & \mu_{1232} & \mu_{1233} \\ \mu_{2111} & \mu_{2112} & \mu_{2113} & \mu_{2121} & \mu_{2122} & \mu_{2123} & \mu_{2131} & \mu_{2132} & \mu_{2133} & \mu_{2121} & \mu_{2122} & \mu_{2123} & \mu_{2131} & \mu_{2132} & \mu_{2133} & \mu_{2231} & \mu_{2232} & \mu_{2233} \\ \mu_{3111} & \mu_{3112} & \mu_{3113} & \mu_{3121} & \mu_{3122} & \mu_{3123} & \mu_{3131} & \mu_{3132} & \mu_{3133} & \mu_{3121} & \mu_{3122} & \mu_{3123} & \mu_{3131} & \mu_{3132} & \mu_{3133} & \mu_{3231} & \mu_{3232} & \mu_{3233} \end{pmatrix}, \quad (8)$$

which should be considered together with the symmetry (3). The tensorial character of the above matrix representation is not of our concern, since all rotations are performed on the fourth-order tensor μ . However, we would like to point out the recent proposal by Le Quang and He [23], providing an explicit matrix representation of the flexoelectric tensor, alternative to the matrix representation by Shu et al. [21] adopted here. Le Quang and He soundly provide an orthogonal tensor transformation that applies to 3×18 matrix representation considering any arbitrary rotation.

2.2. Crystallographic symmetries

In order to perform the Voigt averaging, we need to compute the Eq. (4) accounting for the crystallographic symmetries between the hands, in particular, for tetragonal and cubic crystallites. This information is available only in part in the literature [21].

The cubic crystal system has three crystallographic point groups belonging to the same Laue class, that is, $432 (\mathcal{O})$, $\bar{4}3m (T_d)$, $m\bar{3}m (\mathcal{O}_h)$; while for the tetragonal system we consider four point groups, $422 (D_4)$, $4mm (C_{4v})$, $\bar{4}2m (D_{2d})$, $4/mmm (D_{4h})$ which encompass the T_1 Laue group. These crystallographic point groups are finite subgroups of the orthogonal group $O(3)$ that satisfy the crystallographic restrictions. If $Q \in O(3)$ is any element of the symmetry group of a flexoelectric tensor, it must satisfy

$$\mu = Q^{\otimes 4} \mu. \quad (9)$$

To prove that these crystallographic point group are the symmetry groups of a flexoelectric tensor, it is necessary and sufficient to show that the generators of each point group satisfy the Eq. (9). In what follows, $R(n, \theta)$ stands for the rotation about the unitary vector n through an angle $\theta \in [0, 2\pi)$.

Next, we consider three generators, so equation (9) becomes

$$\mu = Q_3^{\otimes 4} Q_2^{\otimes 4} Q_1^{\otimes 4} \mu. \quad (10)$$

For a cubic system we have that $Q_1 = R(e_3, \pi)$, $Q_2 = R(n, \frac{2\pi}{3})$ with $n = \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3)$, and $Q_3 = R(e_3, \frac{\pi}{2})$ are the generators of $\mathcal{O}$. For T_d , Q_3 is replaced by $Q'_3 = IR(e_3, \frac{\pi}{2})$ where $I = -I$ ($I_{ij} = \delta_{ij}$, being δ_{ij} the Kronecker delta) is the inversion; and for $\mathcal{O}_h$ the generators are Q_1, Q_2, Q_3 and I . Since flexoelectric tensor is fourth-order, inversion operator acts like the identity, that is, $(IR)^{\otimes 4} = (-1)^4 (IR)^{\otimes 4} = R^{\otimes 4}$; so all these point groups represent the same symmetry group. By enforcing the equality (10) it can be shown that the direct flexoelectric tensor μ for cubic materials has three independent coefficients satisfying

$$\begin{aligned} \mu_{1111} &= \mu_{2222} = \mu_{3333} = \mu_{11}, \\ \mu_{1122} &= \mu_{1133} = \mu_{2121} = \mu_{2233} = \mu_{3131} = \mu_{3232} = \mu_{111}, \\ \mu_{1221} &= \mu_{1331} = \mu_{2112} = \mu_{2332} = \mu_{3113} = \mu_{3223} = \mu_{14}. \end{aligned} \quad (11)$$

Note that this result is in accordance with Ref. [21]. Coefficients μ_{11} , μ_{14} and μ_{111} are usually called longitudinal, transverse, and shear coefficients, respectively. This is because the longitudinal flexoelectric coefficient relates the polarization in each direction with the gradient of the longitudinal strain in the same direction. The transverse coefficient relates the polarization and gradient in

the same direction, but the longitudinal strain is taken in the transverse (perpendicular) direction. Instead, shear coefficient relates polarization and gradient of the shear strain in the same direction, as can be deduced from Eq. (2).

For the tetragonal system, the point group D_4 has the rotations $\mathbf{Q}_1 = \mathbf{R}(e_3, \pi)$, $\mathbf{Q}_2 = \mathbf{R}(e_3, \frac{\pi}{2})$ and $\mathbf{Q}_3 = \mathbf{R}(e_1, \pi)$ as generators. For C_{4v} , $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are the same that before, and $\mathbf{Q}_3 = \mathbf{I}\mathbf{R}(e_1, \pi)$. The group D_{2d} is generated by $\mathbf{Q}_1 = \mathbf{R}(e_3, \pi)$, $\mathbf{Q}_2 = \mathbf{R}(e_1, \pi)$ and $\mathbf{Q}_3 = \mathbf{I}\mathbf{R}(e_3, \frac{\pi}{2})$. Last, D_{4h} is generated by the same rotations that D_4 and the inversion $\mathbf{I}$. Like in the previous case, it can be shown that the four crystallographic point groups are representatives of the same symmetry class. Then, any of them leads to the same matrix structure. Once again, by enforcing the equality (10) for these generators, the direct flexoelectric tensor μ for tetragonal materials has eight independent coefficients satisfying

$$\begin{aligned}\mu_{2112} &= \mu_{1221} = \mu_{14}, \\ \mu_{2121} &= \mu_{1122} = \mu_{111}, \\ \mu_{2222} &= \mu_{1111} = \mu_{11}, \\ \mu_{2233} &= \mu_{1133} = \mu_{115}, \\ \mu_{2332} &= \mu_{1331} = \mu_{17}, \\ \mu_{3223} &= \mu_{3113} = \mu_{33}, \\ \mu_{3232} &= \mu_{3131} = \mu_{313}, \\ \mu_{3333} &= \mu_{39}.\end{aligned}\tag{12}$$

This case has not been considered by Shu et al. [21], and to the best of our knowledge, by no one else. However, Le Quang and He [23] present an explicit matrix representation corresponding to their own mapping.

By direct inspection, it can be shown that a direct flexoelectric tensor satisfying

$$\mu_{111} = \frac{\mu_{11} - \mu_{14}}{2},\tag{13}$$

in addition to the conditions (11), it is invariant under any rotation and reflection, that is, the flexoelectric behavior is isotropic. Then, only two coefficients are independent. The isotropy condition (13) differs from that obtained by Shu et al. [21] due to an algebraic mistake.

2.3. Orientation averaging using Wigner-D series expansions

In the following we introduce the generalized spherical harmonic method [15]. Consider a polycrystalline dielectric material that contains individual crystallites with well-characterized flexoelectric properties. In this work, we assume that all crystallites have the same chemical composition. The orientation of each crystallite is defined by a rotation $\mathbf{R}_\beta$ from its initial orientation with respect to a reference, the fixed global coordinate system x_i ($i = 1, 2, 3$). Then, the property for each individual crystallite in the global coordinate system is obtained by means of Eq. (4). The statistical dispersion of the orientation of the crystallites in the sample can be mathematically described by an orientation distribution function (ODF), defined as $w : SO(3) \rightarrow \mathbb{R}_{\geq 0}$, such that [22]

$$w(\mathbf{R}) = \frac{d\wp}{d\mathcal{V}}(\mathbf{R}) \quad \text{for each } \mathbf{R} \in SO(3),\tag{14}$$

where $\wp$ is the orientation measure and $\mathcal{V}$ is the volume measure of $SO(3)$, satisfying the normalization condition

$$\int_{SO(3)} w(\mathbf{R}) d\mathcal{V} = 1.\tag{15}$$

Physically, $w(\mathbf{R})$ describes the probability density of finding $\mathbf{R}$ to be orientation for the single crystals in the sample. This work proposes to estimate the effective direct flexoelectric tensor for polycrystals with an arbitrary distribution of the crystallites, $\mu^V(w)$, through the orientation average of μ weighted by the ODF, that is

$$\mu^V(w) = \langle \mu(\mathbf{R}) \rangle := 8\pi^2 \int_{SO(3)} \mathbf{R}^{\otimes 4} \mu' w(\mathbf{R}) d\mathbf{g}(\mathbf{R}),\tag{16}$$

where $d\mathbf{g}$ is the Haar measure of $SO(3)$, and $8\pi^2 d\mathbf{g} = d\mathcal{V}$. We recall that μ' denotes the direct flexoelectric properties of the single crystal referred to a principal material system. From Eq. (4), we can write

$$\langle \mu_{ijkl} \rangle = \langle R_{ipjqkrls}^{\otimes 4} \mu'_{pqrs} \rangle = \langle R_{ipjqkrls}^{\otimes 4} \rangle \mu'_{pqrs}.\tag{17}$$

Then, in order to obtain the Voigt average for the flexoelectric tensor we should compute the averages for each component of $\mathbf{R}^{\otimes 4}$. Subsequently, we introduce the theory used to accomplish this, then, in the following section we apply it to obtain the closed-form solutions for the averages of cubic and tetragonal crystals.

We introduce the Hilbert space $L^2(SO(3))$, which encompasses the set of all square-integrable complex-valued functions defined on $SO(3)$. This set is endowed with an inner product, defined as follows:

$$(f_1, f_2) = \int_{SO(3)} f_1(g) \overline{f_2(g)} dg, \quad (18)$$

where f_1 and f_2 are elements of $L^2(SO(3))$, and $\bar{z}$ denotes the complex conjugate of z . From the compact Lie group representation theory and by applying the Peter-Weyl theorem, it can be shown that the set $\{\sqrt{2l+1}D_{mn}^l\}$, with D_{mn}^l being the Wigner D-functions, is an orthonormal basis of $L^2(SO(3))$. The Wigner-D functions $D_{mn}^l : SO(3) \rightarrow \mathbb{C}$ with $l = 0, 1, \dots; -l \leq m, n \leq l$ are defined by

$$D_{mn}^l(\mathbf{R}(\psi, \theta, \phi)) = e^{-im\psi} d_{mn}^l(\theta) e^{-in\phi}, \quad (19)$$

where

$$\begin{aligned} d_{mn}^l(\theta) &= \sum_{k=\max\{m-n, 0\}}^{\min\{l-n, l+m\}} \frac{(-1)^k \sqrt{(l+m)!(l-m)!(l+n)!(l-n)!}}{k!(l-n-k)!(l+m-k)!(k-m+n)!} \\ &\times \left(\cos \frac{\theta}{2}\right)^{2l-n+m-2k} \left(\sin \frac{\theta}{2}\right)^{2k-m+n}, \end{aligned} \quad (20)$$

$i = \sqrt{-1}$, and rotations are described by the Euler angles (ψ, θ, ϕ) . Among the various possibilities, we adopt Roe's definition [24] for the Euler angles, so $dV = \sin \theta d\psi d\theta d\phi$. Then, any function κ belonging to $L^2(SO(3))$ admits a Wigner-D series expansion

$$\kappa(\psi, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \sum_{n=-l}^l K_{lmn} d_{mn}^l(\theta) e^{-im\psi} e^{-in\phi}, \quad (21)$$

where K_{lmn} are the expansion coefficients given by

$$K_{lmn} = \frac{2l+1}{8\pi^2} \int_0^{2\pi} \int_0^{\pi} \int_0^{2\pi} \kappa(\psi, \theta, \phi) d_{mn}^l(\theta) e^{im\psi} e^{in\phi} d\psi d\theta d\phi, \quad (22)$$

this is, the inner product between κ and $(2l+1)D_{mn}^l$ [25]. In particular, for the ODF w we can write

$$w(\psi, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \sum_{n=-l}^l c_{mn}^l d_{mn}^l(\theta) e^{-im\psi} e^{-in\phi}, \quad (23)$$

where the expansion coefficients c_{mn}^l are given by

$$c_{mn}^l = \frac{2l+1}{8\pi^2} \int_0^{2\pi} \int_0^{\pi} \int_0^{2\pi} w(\psi, \theta, \phi) d_{mn}^l(\theta) e^{im\psi} e^{in\phi} d\psi d\theta d\phi. \quad (24)$$

Due to the symmetry properties of d_{mn}^l [22], and since w is a real-valued function, the texture coefficients satisfy

$$c_{\bar{m}\bar{n}}^l = (-1)^{n-m} \overline{c_{mn}^l}, \quad (25)$$

where $\bar{m} = -m$. Due to the normalization condition (15) and the properties of the Wigner-D functions

$$c_{00}^0 = \frac{1}{8\pi^2}. \quad (26)$$

The coefficients c_{mn}^l ($m, n, l \neq 0$), known as *texture coefficients*, encapsulate the entirety of the texture information, as delineated in Eq. (24). In the case that all texture coefficients become zero, the ODF simplifies to $w_{\text{iso}} = \frac{1}{8\pi^2}$, as can be observed from expansion (23). This scenario corresponds to a completely random distribution of crystallites, thereby resulting in an isotropic behavior.

Consider the Voigt average of a general function, $\kappa \in L^2(SO(3))$,

$$\langle \kappa \rangle = 8\pi^2 \int_{SO(3)} \kappa(g) w(g) dg. \quad (27)$$

Because the ODF is a real-valued function, the above average can be computed from the inner product, $\langle \kappa \rangle = 8\pi^2(\kappa, w)$. Then, by replacing the expansions (21) and (23) in this inner product we obtain [15]

$$\langle \kappa \rangle = 8\pi^2 \sum_{l=0}^{\infty} \sum_{m=-l}^l \sum_{n=-l}^l \frac{1}{2l+1} K_{lmn} c_{mn}^l. \quad (28)$$

3. Formulas for flexoelectricities

This section introduces closed-form expressions for all the components of the flexoelectricity matrix (8) in term of the properties of the crystallites and the texture coefficients.

First, note that each component of $\mathbf{R}^{\otimes 4}$ is a real function defined over $SO(3)$. Then, Eq. (28) can be used to compute the Voigt average of $\mathbf{R}^{\otimes 4}$, and after Eq. (17), the average of the flexoelectric tensor. By virtue of the symmetry of the flexoelectric tensor μ' , for both the case of tetragonal and cubic crystallites, the amount of different $R_{ijklmnop}^{\otimes 4}$ involved in (17) is significantly reduced from the 1545 mentioned in Section 2.1 to 396, with the subsequent reduction of the computational cost. However, it is worth clarifying that the appeal of this formulation is that we need to compute the expansion coefficients only the first time, and then the analytic solution is always available.

From the theorem by Ferrari and Johnson [26], the expansion coefficients of the components of a matrix transformation between tensors of order r vanish for all $l > r$. Therefore, when κ in (28) is one of the components of $\mathbf{R}^{\otimes 4}$, its average results in [15]

$$\langle R_{ijkopqrs}^{\otimes 4} \rangle = 8\pi^2 \sum_{l=0}^4 \sum_{m=-l}^l \sum_{n=-l}^l \frac{1}{2l+1} K_{lmn} c_{mn}^l, \quad (29)$$

where the expansion coefficients K_{lmn} ($l \leq 4$; $-l \leq m, n \leq l$) are computed as in (22) but using the corresponding component of $\mathbf{R}^{\otimes 4}$ as kernel. The integration has been done analytically by using Mathematica [27]. Once we have obtained the averages (29), in order to obtain the Voigt average of the flexoelectric tensor, we have two possible paths: either (i) to impose in the series solution the restrictions on the texture coefficients according to the crystallographic symmetry [22], or (ii) to implement the theory developed by Buroni and Buroni [15], that is, to use in (17) the matrix structure for μ' as given by (11) or (12), accordingly to the crystallographic symmetry of the crystallite. We have adopted the last direct method. Due to space limitations, the 54 expressions for each of the flexoelectric coefficients for polycrystals with cubic and tetragonal crystallites are reported in the Supplementary Material.

Crystal symmetry imposes restrictions on the n index of texture coefficients, while texture symmetry imposes restrictions on the m index. In the case of even-order tensors, such as the flexoelectric, and crystals with cubic symmetry, the texture coefficients $c_{mn}^l = 0$ when $n \neq 4k$, $k \in \mathbb{Z}$. If $G_{\text{tex}} \subset SO(3)$ is the group of texture symmetries, the ODF must remain invariant under all elements of this group. For a fiber texture of type C_∞ , the group of texture symmetries is generated by a rotation of any angle about an axis. Taking the rotation axis as x_3 , this imposes the following additional restrictions on the textures coefficients [22]

$$c_{mn}^l \neq 0 \Rightarrow m = 0. \quad (30)$$

It is also known that an ODF of this type is independent of the angle ψ [28]. Then, the non-zero Voigt averaged coefficients for fiber texture and cubic crystals reduce to:

$$\begin{aligned} \langle \mu_{1111} \rangle &= \frac{2}{105} \pi^2 \left(84c_{00}^0 \mu_B - \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{1122} \rangle &= -\frac{2}{315} \pi^2 \left(252c_{00}^0 \mu_C + \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{1133} \rangle &= \frac{8}{315} \pi^2 \left(-63c_{00}^0 \mu_C + \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{1221} \rangle &= \frac{2}{315} \pi^2 \left(252c_{00}^0 (\mu_B + 2\mu_C) - \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{1331} \rangle &= \frac{8}{315} \pi^2 \left(63c_{00}^0 (\mu_B + 2\mu_C) + \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{3333} \rangle &= \frac{8}{315} \pi^2 \left(63c_{00}^0 \mu_B - 2 \text{Re} \left(7c_{00}^4 + \sqrt{70}c_{04}^4 \right) \mu_A \right), \\ \langle \mu_{2222} \rangle &= \langle \mu_{1111} \rangle, \\ \langle \mu_{2121} \rangle &= \langle \mu_{1122} \rangle, \\ \langle \mu_{2233} \rangle &= \langle \mu_{3131} \rangle = \langle \mu_{3232} \rangle = \langle \mu_{1133} \rangle, \\ \langle \mu_{2112} \rangle &= \langle \mu_{1221} \rangle, \\ \langle \mu_{2332} \rangle &= \langle \mu_{3113} \rangle = \langle \mu_{3223} \rangle = \langle \mu_{1331} \rangle, \end{aligned} \quad (31)$$

where $\mu_A = \mu_{2112} + 2\mu_{2121} - \mu_{2222}$, $\mu_B = 2\mu_{2112} + 4\mu_{2121} + 3\mu_{2222}$, $\mu_C = \mu_{2112} - 3\mu_{2121} - \mu_{2222}$ and $c_{00}^0 = \frac{1}{8\pi^2}$. $\text{Re}(z)$ stands for the real part of z .

Our final remark highlights several relevant points regarding the Voigt-type model. Consider a polycrystalline aggregate with constitutive law (2) for each point $\mathbf{x}$ of the domain. Then, by averaging, we have

$$\overline{P_i(\mathbf{x})} = \overline{\mu_{ijkl}(\mathbf{x}) \frac{\partial S_{jk}(\mathbf{x})}{\partial x_l}} = \overline{\mu_{ijkl}(\mathbf{x})} \frac{\partial \overline{S_{jk}}}{\partial x_l}, \quad (32)$$

recalling that overbar denotes the volume average, i.e., $\overline{(\cdot)} := \frac{1}{V} \int_V (\cdot) dV$ with V being the volume, and $\mu_{ijkl}(\mathbf{x})$ depends on the orientation of the individual crystallite located at $\mathbf{x}$. The second equality holds true if, and only if, the gradient of the strains is constant throughout the aggregate, which allows us to factor out $\frac{\partial S_{jk}}{\partial x_l}$ from the averaging integral. The averages $\overline{P_i}$ and $\frac{\partial \overline{S_{jk}}}{\partial x_l}$ are identified with the effective electric polarization vector and strain gradient, respectively. Consequently, we observe that the effective

Table 2
Experimental and calculated values of direct flexoelectric coefficients for BTO.

μ_{11} (nC/m)			μ_{14} (nC/m)			μ_{111} (nC/m)		
Experiment [29]	DFT [7]	LDA [6]	Experiment [29]	DFT [7]	LDA [6]	Experiment [29]	DFT [7]	LDA [6]
5×10^4	-0.36	0.15	5×10^4	1.6	1.904	—	-1.5	-5.46

direct flexoelectric tensor in this model is approximated by $\overline{\mu(\mathbf{x})}$. In this study, the volume average $\overline{\mu(\mathbf{x})}$ has been computed through the orientation average $\langle \mu(\mathbf{R}) \rangle$.

It is well-known that the analogous problem was studied by Voigt in 1887 in the context of elasticity. In that case, or even for piezoelectricity, the estimation for the effective elastic stiffness tensor is calculated by averaging the microscopic distribution of properties, assuming that strain fields are uniform within the aggregate. Note, however, that there is no such an assumption here. Instead, uniform strain gradient fields are assumed, as mentioned above.

The Voigt-type model can be viewed as a first-order approximation that primarily uses the statistics of the orientation distribution, represented as a one-point correlation function (the ODF). This model can be enhanced by incorporating more detailed information about the statistics of the microstructure, such as higher-order correlation functions; and/or by relaxing the model assumption about the strain gradient fields. In this context, having closed-form expressions for this first-order approximation is particularly valuable as it provides a straightforward and computationally efficient way to compute higher-order estimations.

4. Results

In order to validate the formulation, we first consider the ODF

$$w_\delta(\psi, \theta, \phi) = \delta(\psi)\delta(\theta)\delta(\phi), \quad (33)$$

where $\delta(\xi)$ is the Dirac delta function at $\xi = 0$. The texture coefficients in this case are:

$$c_{mn}^{\delta,l} = \begin{cases} \frac{2l+1}{8\pi^2}, & \text{if } m = n \\ 0, & \text{if } m \neq n. \end{cases} \quad (34)$$

Explicitly, the nonzero coefficients that appear in the average series for a fourth-order tensor are

$$c_{00}^{\delta,0} = \frac{1}{8\pi^2}, \quad (35a)$$

$$c_{11}^{\delta,1} = c_{00}^{\delta,1} = c_{11}^{\delta,1} = \frac{3}{8\pi^2}, \quad (35b)$$

$$c_{22}^{\delta,2} = c_{11}^{\delta,2} = c_{00}^{\delta,2} = c_{11}^{\delta,2} = c_{22}^{\delta,2} = \frac{5}{8\pi^2}, \quad (35c)$$

$$c_{33}^{\delta,3} = c_{22}^{\delta,3} = c_{11}^{\delta,3} = c_{00}^{\delta,3} = c_{11}^{\delta,3} = c_{22}^{\delta,3} = c_{33}^{\delta,3} = \frac{7}{8\pi^2}, \quad (35d)$$

$$c_{44}^{\delta,4} = c_{33}^{\delta,4} = c_{22}^{\delta,4} = c_{11}^{\delta,4} = c_{00}^{\delta,4} = c_{11}^{\delta,4} = c_{22}^{\delta,4} = c_{33}^{\delta,4} = c_{44}^{\delta,4} = \frac{9}{8\pi^2}. \quad (35e)$$

The ODF (33) corresponds to the situation in which there is no possibility that any crystallite is oriented differently to $\psi = \theta = \phi = 0$. Then, the single-crystal properties should be recovered from the effective properties of the polycrystal. This is successfully verified by substituting the texture coefficients (35) into the expressions for the average of cubic and tetragonal crystallites reported in Supplementary Material.

Moreover, it has been verified that tetragonal solutions reduce to the cubic ones when evaluated for cubic materials, as expected. Unfortunately, at present, tetragonal properties have not been well characterized. In the next section, we perform some computations for cubic crystals.

4.1. Crystallites with cubic symmetry

The flexoelectric properties for the BTO are available in the literature only for its (cubic) paraelectric phase. The experimental and numerical values of the flexoelectric coefficients for BTO with cubic symmetry have been taken from Maranganti and Sharma [6], and the recent review by Shu et al. [29]. Maranganti and Sharma [6] have estimated the properties via the lattice dynamics approach (LDA). Additionally, numerical values calculated by Xu et al. [7] through first-principles calculations based on the density functional theory (DFT) are also considered. All these properties are summarized in Table 2. The order-of-magnitude discrepancy between experimental and theoretical results of flexoelectric coefficients is well-known and its cause is currently under debate. It is suspected that some extrinsic contributions are unintentionally mixed with intrinsic flexoelectricity in experimental measurement, or that some key features of flexoelectricity have not been appropriately considered by theoretical models. This issue is discussed with more detail in Ref. [1]. In this work, we select the numerically obtained properties for presenting our results.

We first consider the scenario of untextured (isotropic) ceramics, which can be easily computed by setting the texture coefficients to zero in Equations (31). The effective flexoelectricities, considering both DFT [7] and LDA [6] estimations for the crystal properties,

Table 3

Estimations of flexoelectric coefficients for isotropic ceramics with crystal properties of BTO computed by DFT and LDA methods.

μ_{11} (nC/m)		μ_{14} (nC/m)		μ_{111} (nC/m)	
DFT [7]	LDA [6]	DFT [7]	LDA [6]	DFT [7]	LDA [6]
-0.776	-3.516	1.808	3.737	-1.292	-3.626

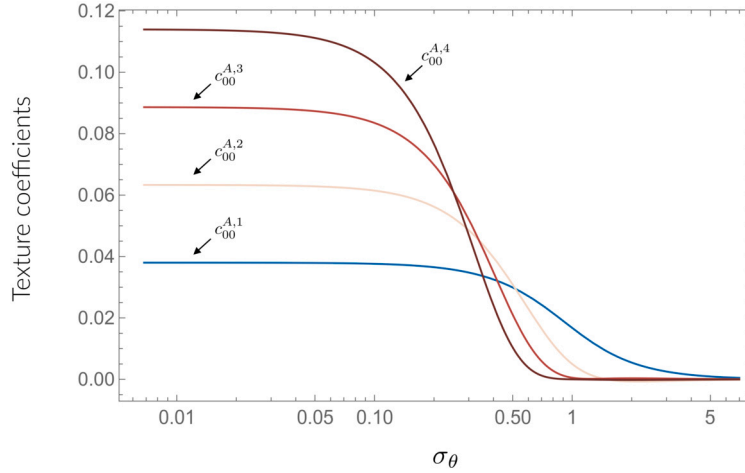


Fig. 2. Texture coefficients $c_{00}^{A,i}$ ($i = 1 \dots 4$) as a function of the Gaussian distribution parameter σ_θ when the mean value γ_θ is zero.

are displayed in Table 3. It is noteworthy that these results exactly fulfill the isotropy condition (13). Ma and Cross [30] have conducted experimental measurements on the unpoled ceramic of BTO in its cubic phase. They discovered that μ_{14} reaches nearly 50 $\mu\text{C}/\text{m}$ around the phase transition point at the Curie Temperature (120°C), after which the flexoelectric coefficient decays to values around 10 $\mu\text{C}/\text{m}$. It is important to note that the discrepancy between these experimental measurements and our estimations is of the same order as those for the single crystals mentioned above. This discrepancy underscores the complexity of the field and the challenges inherent in measuring and modeling flexoelectric properties, a topic that is currently under debate [1].

Now, we consider a Gaussian fiber texture given by

$$w_A(\theta | \gamma_\theta, \sigma_\theta) = \frac{\exp\left(-\frac{(\theta - \gamma_\theta)^2}{2\sigma_\theta^2}\right)}{\sqrt{2\pi}\sigma_\theta}, \quad (36)$$

where γ_θ is the mean of the Euler angle θ , and σ_θ is its Gaussian standard deviation. The orientations due to angles ψ and ϕ are assumed to be randomly distributed. The ODF w_A is used to compute the texture coefficients through equation (24). This results in the non-zero texture coefficients, $c_{00}^{A,1}$, $c_{00}^{A,2}$, $c_{00}^{A,3}$ and $c_{00}^{A,4}$, which are all real-valued functions. Fig. 2 shows the evolution of these texture coefficients as a function of the parameter σ_θ when the mean value γ_θ is zero. It is observed that when $\sigma_\theta \rightarrow 0$ the texture coefficients tend to their values corresponding to a delta Dirac distribution, that is, the values in equation (35), as expected. In contrast, the situation where the Euler angle θ is uniformly distributed is modeled as $\sigma_\theta \rightarrow \infty$, in which case it can be observed that all texture coefficients vanish, and the resulting behavior is isotropic.

Consider a cubic crystallite with properties taken from Ref. [6] as indicated in Table 2. The ODF w_A is a particular case of fiber texture where the angle ϕ is uniformly distributed. Then, the Voigt averages can be obtained from the Eqs. (31), from where become clear that the average depends only on the texture coefficients $c_{00}^{A,4}$ and $c_{04}^{A,4}$. For any ODF that is independent of ϕ , it can be seen from Eq. (24) that $c_{mn}^I = 0$ when $n = 4k$, $k \in \mathbb{Z} - \{0\}$. In particular, $c_{04}^{A,4} = 0$, as already mentioned. As a consequence, the Voigt average only depends on the texture coefficient $c_{00}^{A,4}$. Fig. 3 shows the absolute value of the Voigt estimation of the flexoelectric coefficients, μ_{11}^V , μ_{111}^V and μ_{14}^V , as a function of the mean value γ_θ . Continuous lines correspond to $\sigma_\theta \rightarrow 0$, while dashed lines correspond to $\sigma_\theta = 5$. It is worth noting that these are not the only non-zero flexoelectric coefficients since the macroscopic behavior is transversely isotropic, not cubic. However, we show only these three constants to get clean plots. The coefficients μ_{11}^V and μ_{111}^V are negative. It can be seen that the flexoelectric coefficients have global maximum and minimum values at $\gamma_\theta = 0$ and $\gamma_\theta \approx 0.857 \approx 49.1^\circ$ when $\sigma_\theta \rightarrow 0$. In contrast, the flexoelectric coefficients do not change with γ_θ when $\sigma_\theta = 5$, that is, the behavior is essentially isotropic. This can be verified since the constants are: $\mu_{11}^V = -3.5173$ nC/m, $\mu_{14}^V = 3.7368$ nC/m, and $\mu_{111}^V = -3.6271$ nC/m, which satisfies the isotropy condition (13), i.e., $\mu_{11}^V = \mu_{14}^V + 2\mu_{111}^V$. Fig. 4 shows the absolute value of the Voigt estimation of the flexoelectric coefficients as a function of the parameter σ_θ . The continuous lines correspond to the ODF with a mean value $\gamma_\theta = 0$, while the dashed lines correspond to the ODF with a mean value $\gamma_\theta \approx 0.857 \approx 49.1^\circ$. The range of variations of the properties when the mean value γ_θ varies between

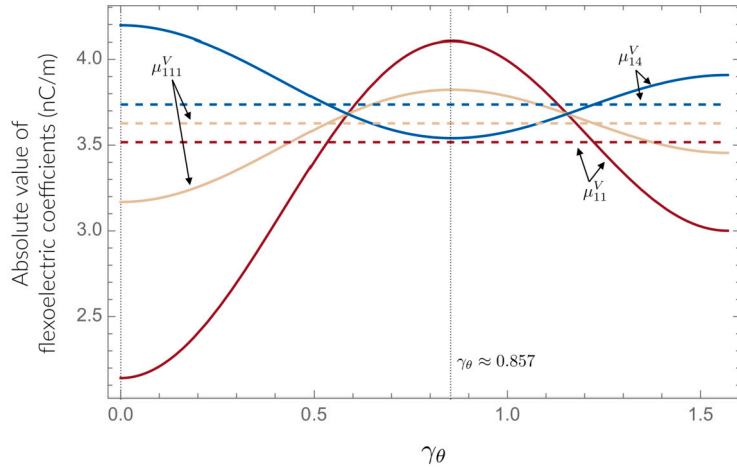


Fig. 3. Absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the mean value γ_θ . Continuous lines correspond to $\sigma_\theta \rightarrow 0$. Dashed lines correspond to $\sigma_\theta = 5$.

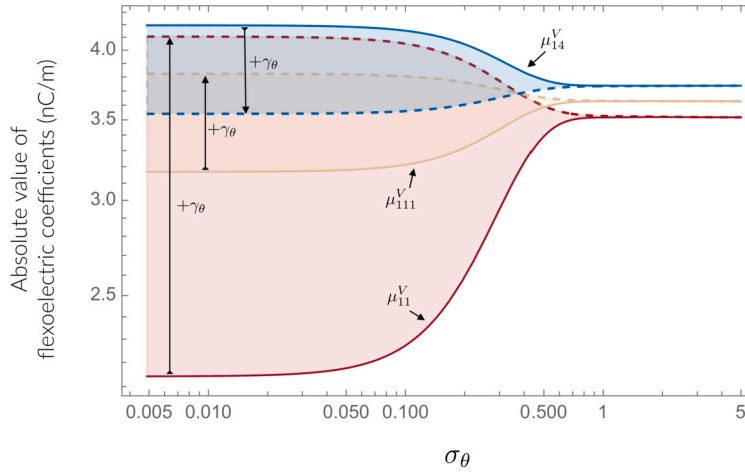


Fig. 4. Absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the Gaussian distribution parameter σ_θ .

these two extremes is also represented. It is observed that, depending on γ_θ , the absolute values of the flexoelectricities can grow or decrease with the standard deviation σ_θ . An interesting observation emerging from our theory is that, unlike the piezoelectric properties [20], nonzero flexoelectric properties can be obtained in a fully random polycrystalline material, as revealed in Fig. 4. This behavior is expected for even-order tensors, such as the direct flexoelectric. In fact, for a dispersion σ_θ approximately greater than 1, the behavior is essentially isotropic. When $\sigma_\theta \rightarrow \infty$ the flexoelectricities become $\mu_{11}^V = -3.5164$ nC/m, $\mu_{14}^V = 3.7372$ nC/m, and $\mu_{111}^V = -3.6268$ nC/m, which satisfies the isotropy condition (13), i.e., $\mu_{11}^V = \mu_{14}^V + 2\mu_{111}^V$. This study appears to be the first to quantify the flexoelectric transition from texture to isotropic behavior.

Fig. 5 shows the absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the Gaussian distribution parameter σ_θ for a mean value $\gamma_\theta = 0$ obtained for crystals with properties computed by Maranganti and Sharma [6] and with properties computed by Xu et al. [7]. It is observed that the overall behavior is the same in both cases.

By arguing that the shear flexoelectric coefficient is not well characterized (see Table 2), in some numerical approaches, where it is assumed that the gradients of the shear strains are marginal, it is a common practice to set the shear flexoelectric coefficient to zero [31–33]. This fact motivates the subsequent study of the effect of the shear flexoelectric coefficient of the crystallite on the effective properties. For this purpose, we propose the following flexoelectric anisotropy index

$$\mathcal{A} := \frac{2\mu_{111}}{\mu_{11} - \mu_{14}}, \quad (37)$$

in such a way that $\mathcal{A} = 1$ means that the material is isotropic. The index $\mathcal{A}$ quantifies the degree of flexoelectric anisotropy in cubic materials. Fig. 6 shows the variation of the absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the parameter $\mathcal{A}$ when the mean value $\gamma_\theta = 0$. Continuous lines correspond to a $\sigma_\theta = 0.1$ while dashed lines correspond to $\sigma_\theta = 5$. The μ_{11} and μ_{14} coefficients for the crystal are taken from Maranganti and Sharma [6] (see Table 2). The shear flexoelectric coefficient is

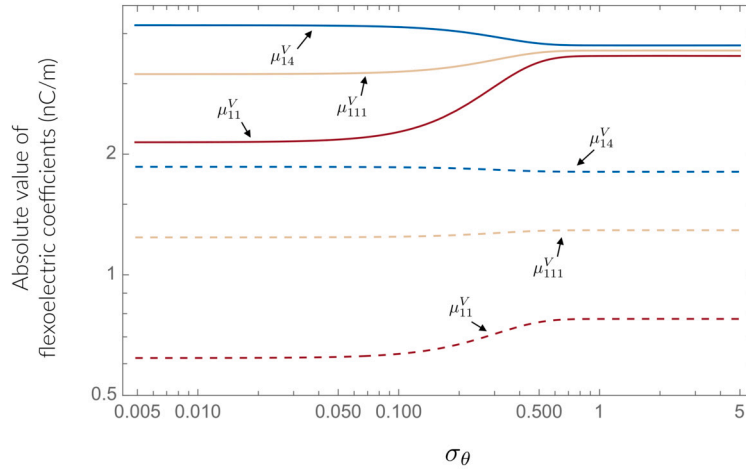


Fig. 5. Absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the Gaussian distribution parameter σ_θ for a mean value $\gamma_\theta = 0$. Continuous lines correspond to crystals properties computed by Maranganti and Sharma [6], while dashed lines are computed by Xu et al. [7].

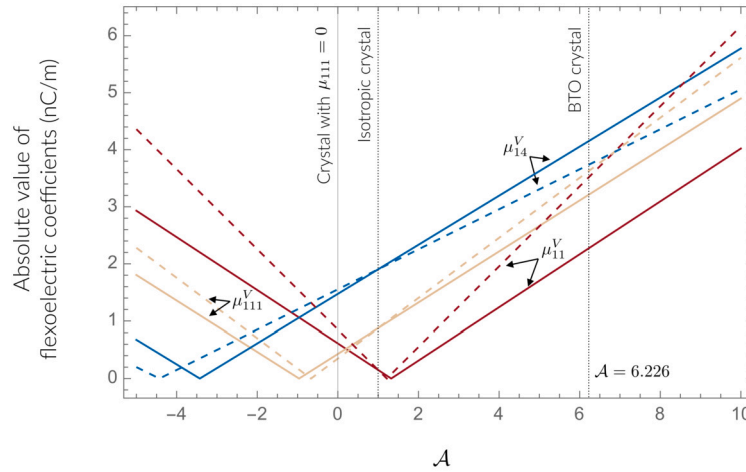


Fig. 6. Absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the parameter $\mathcal{A}$. The mean value $\gamma_\theta = 0$. Continuous lines correspond to a $\sigma_\theta = 0.1$. Dashed lines correspond to $\sigma_\theta = 5$.

taken as $\mu_{111} = \mathcal{A} \frac{\mu_{11} - \mu_{14}}{2}$. The effective properties for the BTO crystals correspond to the value of the flexoelectric anisotropy index $\mathcal{A} \approx 6.226$. A decrease in the effective flexoelectric coupling is observed when the crystal is assumed isotropic ($\mathcal{A} = 1$) or with a zero shear flexoelectric coefficient ($\mathcal{A} = 0$). In both cases, a signal change is observed in the component μ_{11}^V .

Next, we consider an aggregate with biaxial Gaussian distributions [34] for the angles θ and ψ , and a random distribution for ϕ given by

$$w_B(\theta | \gamma_\theta, \sigma_\theta; \psi | \gamma_\psi, \sigma_\psi) = \frac{\exp\left(-\frac{(\theta - \gamma_\theta)^2}{2\sigma_\theta^2} - \frac{(\psi - \gamma_\psi)^2}{2\sigma_\psi^2}\right)}{2\pi\sigma_\theta\sigma_\psi}, \quad (38)$$

where γ_θ and γ_ψ are the mean values for the Euler angles, θ and ψ ; and σ_θ and σ_ψ the corresponding standard deviations. In this case, there are 24 non-zero texture coefficients with $l \leq 4$, which are complex. The evolution of the texture coefficients $c_{mn}^{B,l}$ as a function of the Gaussian distribution parameter σ_θ is given in Fig. 7. Arbitrarily, the mean values are $\gamma_\theta = 0$ and $\gamma_\psi = \frac{\pi}{3}$; and the dispersion of ψ is $\sigma_\psi = 1$. It is observed that some of the texture coefficients vanish for standard deviations tending to infinity or tending to zero, while others converge to finite values. In particular, the imaginary parts of all the complex coefficients vanish for $\sigma_\theta \rightarrow 0$, while only the texture coefficients $c_{00}^{B,1}$, $c_{00}^{B,2}$, $c_{00}^{B,3}$ and $c_{00}^{B,4}$ take finite values. In fact, the real part of these coefficients matches exactly with $c_{00}^{A,1}$, $c_{00}^{A,2}$, $c_{00}^{A,3}$ and $c_{00}^{A,4}$ for all σ_θ , respectively (see Fig. 2). As an illustrative example of the overall behavior of the polycrystal with a distribution given by w_B , consider the components μ_{11}^V , μ_{22}^V and μ_{33}^V . Despite the fact that in cubic crystals we have $\mu_{11} = \mu_{22} = \mu_{33}$, the Voigt averages are clearly different, as shown in Fig. 8. In comparison, in the previous example where the fiber texture given by w_A is considered, the Voigt averages satisfy $\mu_{11}^V = \mu_{22}^V \neq \mu_{33}^V$, in concordance with Eq. (31).

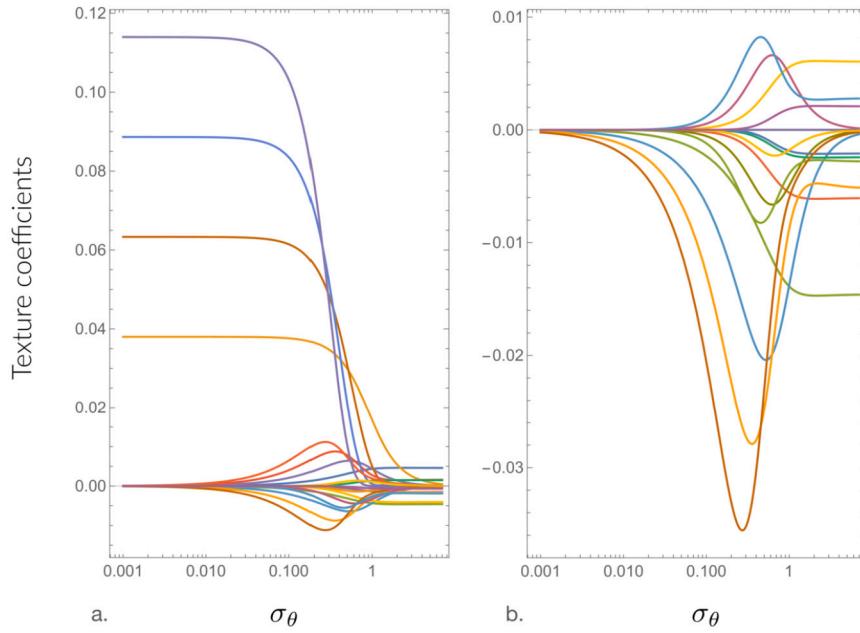


Fig. 7. The 24 non-zero texture coefficients $c_{mn}^{B,I}$ as a function of the Gaussian distribution parameter σ_θ when the mean values are $\gamma_\theta = 0$ and $\gamma_\psi = \frac{\pi}{3}$; and $\sigma_\psi = 1$. (a) Real parts. (b) Imaginary parts.

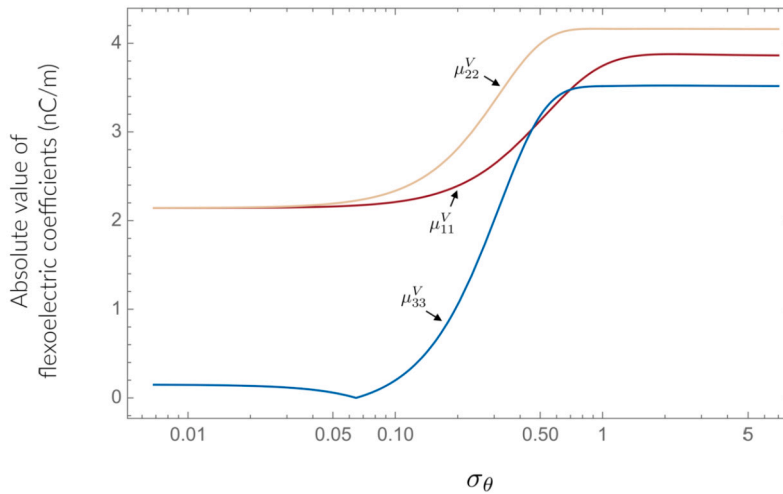


Fig. 8. Absolute value of the Voigt estimate of the flexoelectric coefficients as a function of the Gaussian distribution parameter σ_θ . Mean values are $\gamma_\theta = 0$ and $\gamma_\psi = \frac{\pi}{3}$; and $\sigma_\psi = 0.1$.

Along the presented examples, we have considered the behavior of only a few components of the flexoelectricities in order to get clean figures. However, since we have deduced expressions for the 54 components of the Voigt averages, this study provides a complete picture of the effective flexoelectric tensors according to the ODF. Thus, Table 4 presents the structure of the 3×18 matrices of the Voigt averages corresponding to the analyzed ODFs, w_A and w_B , showing which components become equal or different from zero. It can be observed that the simple Gaussian ODF w_A results in a material with an effective flexoelectricity tensor with cubic symmetry class and fiber texture, whose components satisfy the equalities of Eq. (31). On the other hand, the biaxial ODF w_B in the angles θ and ψ produce a fully-populated effective flexoelectric tensor with 18 independent coefficients, and the following equalities are satisfied

Table 4

Matrix structures of the effective flexoelectric tensors resulting from the two Gaussian ODFs.

Crystal matrix structure																	
$\begin{pmatrix} \mu_{1111} & 0 & 0 & \mu_{1221} & 0 & 0 & \mu_{1221} & 0 & 0 & 0 & \mu_{1122} & 0 & 0 & 0 & \mu_{1122} & 0 & 0 & 0 \\ 0 & \mu_{1221} & 0 & 0 & \mu_{1111} & 0 & 0 & \mu_{1221} & 0 & \mu_{1122} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mu_{1122} \\ 0 & 0 & \mu_{1221} & 0 & 0 & \mu_{1221} & 0 & 0 & \mu_{1111} & 0 & 0 & 0 & \mu_{1122} & 0 & 0 & 0 & \mu_{1122} & 0 \end{pmatrix}$																	
ODF	Polycrystal matrix structure																
w_A	$\begin{pmatrix} \langle \mu_{1111} \rangle & 0 & 0 & \langle \mu_{1221} \rangle & 0 & 0 & \langle \mu_{1331} \rangle & 0 & 0 & 0 & \langle \mu_{1122} \rangle & 0 & 0 & 0 & \langle \mu_{1133} \rangle & 0 & 0 & 0 \\ 0 & \langle \mu_{1221} \rangle & 0 & 0 & \langle \mu_{1111} \rangle & 0 & 0 & \langle \mu_{1331} \rangle & 0 & \langle \mu_{1122} \rangle & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \langle \mu_{1133} \rangle \\ 0 & 0 & \langle \mu_{1331} \rangle & 0 & 0 & \langle \mu_{1331} \rangle & 0 & 0 & \langle \mu_{3333} \rangle & 0 & 0 & 0 & \langle \mu_{1133} \rangle & 0 & 0 & 0 & \langle \mu_{1133} \rangle & 0 \end{pmatrix}$																
w_B	$\begin{pmatrix} \langle \mu_{1111} \rangle & \langle \mu_{1112} \rangle & \langle \mu_{1113} \rangle & \langle \mu_{1221} \rangle & \langle \mu_{1222} \rangle & \langle \mu_{1223} \rangle & \langle \mu_{1331} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{1333} \rangle & \langle \mu_{1112} \rangle & \langle \mu_{1122} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1113} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1133} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1223} \rangle & \langle \mu_{1332} \rangle \\ \langle \mu_{1112} \rangle & \langle \mu_{1221} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1222} \rangle & \langle \mu_{2222} \rangle & \langle \mu_{2223} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{2332} \rangle & \langle \mu_{2333} \rangle & \langle \mu_{1122} \rangle & \langle \mu_{1222} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1133} \rangle & \langle \mu_{1232} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{1223} \rangle & \langle \mu_{2223} \rangle & \langle \mu_{2333} \rangle \\ \langle \mu_{1113} \rangle & \langle \mu_{1122} \rangle & \langle \mu_{1331} \rangle & \langle \mu_{1223} \rangle & \langle \mu_{2223} \rangle & \langle \mu_{2332} \rangle & \langle \mu_{1333} \rangle & \langle \mu_{2333} \rangle & \langle \mu_{3333} \rangle & \langle \mu_{1123} \rangle & \langle \mu_{1223} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{1133} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{1333} \rangle & \langle \mu_{1332} \rangle & \langle \mu_{2233} \rangle & \langle \mu_{2333} \rangle \end{pmatrix}$																

$$\begin{aligned}
\langle \mu_{1121} \rangle &= \langle \mu_{2111} \rangle = \langle \mu_{1112} \rangle = \langle \mu_{12} \rangle, \\
\langle \mu_{1131} \rangle &= \langle \mu_{3111} \rangle = \langle \mu_{1113} \rangle = \langle \mu_{13} \rangle, \\
\langle \mu_{2112} \rangle &= \langle \mu_{1221} \rangle = \langle \mu_{14} \rangle, \\
\langle \mu_{2122} \rangle &= \langle \mu_{2221} \rangle = \langle \mu_{1222} \rangle = \langle \mu_{15} \rangle, \\
\langle \mu_{1232} \rangle &= \langle \mu_{2123} \rangle = \langle \mu_{2132} \rangle = \langle \mu_{2231} \rangle = \langle \mu_{3122} \rangle = \langle \mu_{3221} \rangle = \langle \mu_{1223} \rangle = \langle \mu_{16} \rangle, \\
\langle \mu_{3113} \rangle &= \langle \mu_{1331} \rangle = \langle \mu_{17} \rangle, \\
\langle \mu_{1233} \rangle &= \langle \mu_{2133} \rangle = \langle \mu_{2331} \rangle = \langle \mu_{3123} \rangle = \langle \mu_{3132} \rangle = \langle \mu_{3231} \rangle = \langle \mu_{1332} \rangle = \langle \mu_{18} \rangle, \\
\langle \mu_{3133} \rangle &= \langle \mu_{3331} \rangle = \langle \mu_{1333} \rangle = \langle \mu_{19} \rangle, \\
\langle \mu_{2121} \rangle &= \langle \mu_{1122} \rangle = \langle \mu_{111} \rangle, \\
\langle \mu_{1132} \rangle &= \langle \mu_{1231} \rangle = \langle \mu_{2113} \rangle = \langle \mu_{2131} \rangle = \langle \mu_{3112} \rangle = \langle \mu_{3121} \rangle = \langle \mu_{1123} \rangle = \langle \mu_{112} \rangle, \\
\langle \mu_{3131} \rangle &= \langle \mu_{1133} \rangle = \langle \mu_{115} \rangle, \\
\langle \mu_{2232} \rangle &= \langle \mu_{3222} \rangle = \langle \mu_{2223} \rangle = \langle \mu_{26} \rangle, \\
\langle \mu_{3223} \rangle &= \langle \mu_{2332} \rangle = \langle \mu_{28} \rangle, \\
\langle \mu_{3233} \rangle &= \langle \mu_{3332} \rangle = \langle \mu_{2333} \rangle = \langle \mu_{29} \rangle, \\
\langle \mu_{3232} \rangle &= \langle \mu_{2233} \rangle = \langle \mu_{218} \rangle.
\end{aligned} \tag{39}$$

5. Conclusions

In this work, we have presented for the first time, closed-form expressions that allow the direct flexoelectric coefficients to be computed for textured polycrystalline dielectrics. These expressions provide a significant first step towards the estimations of the effective flexoelectricities in aggregates, in this case, a first-order estimation given by the Voigt average. Crystallites can belong to any class of cubic symmetry (i.e. 432 ($\mathcal{O}$), $\bar{4}3m$ (T_d), or $m\bar{3}m$ ($\mathcal{O}_h$)), or T_1 tetragonal symmetry (i.e. 422 (D_4), $4mm$ (C_{4v}), $\bar{4}2m$ (D_{2d}), or $4/mmm$ (D_{4h})); while no restriction is imposed on texture symmetry. The methodology is leveraged into the orthogonality properties of the Wigner-D functions. In the derivation, we have exploited the symmetries of the flexoelectric tensor for cubic and tetragonal crystals, and the symmetries of the quadruple tensor product of the proper orthogonal matrix, which have been described in detail. The expressions reduced to the case of cubic crystallographic symmetry with fiber texture symmetry have also been presented. Additionally, the correct isotropy condition has been derived after revisiting previous results.

In light of our exploration, the Voigt-type model emerges as an essential first-order approximation, leveraging the statistics of the orientation distribution encapsulated in the one-point correlation function (the ODF). This model, while simple, provides a strong foundation for more detailed statistical analyses of the microstructure and strain gradient fields. The closed-form expressions derived in this study are particularly useful, as they offer a computationally efficient pathway to higher-order estimations. These expressions, therefore, serve as a compelling starting point for homogenization relations based on perturbation expansions, highlighting the significance of our findings in the broader context of material science research. However, it is important to note that while our closed-form expressions provide a valuable tool for estimating the effective flexoelectric coefficients of textured polycrystalline dielectrics, we acknowledge that further research is needed for comprehensive validation. This includes comparison with more extensive experimental data and realistic simulations.

Note that the derived closed-form expressions are also applicable to the fourth-order flexomagnetic tensor.

CRediT authorship contribution statement

JLB: Methodology, Investigation, Software, Validation, Writing – Original Draft, Writing – Review & Editing. RM: Writing – Review & Editing. LRT: Project Administration. AS: Writing – Review & Editing. FCB: Conceptualization, Methodology, Formal Analysis, Investigation, Software, Visualization, Writing – Original Draft, Writing - Review & Editing, Project Administration, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

Financial support for this research was provided by the Consejería de Economía, Conocimiento, Empresas y Universidad de la Junta de Andalucía (Spain) through the research project P18-RT-3128, which has been co-funded by the Fondo Europeo de Desarrollo Regional (FEDER), Programa Operativo FEDER 2014–2020. Additionally, the authors gratefully acknowledge the financial support provided by the Ministerio de Ciencia e Innovación (Spain) through the research project PID2022-137903OB-I00. R.M. is also acknowledging support of NSERC and CRC Program.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.apm.2023.09.032>.

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